Evaluating Driver Satisfaction with Autonomous Braking Behavior in Simulated Driving Conditions

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Abstract

This paper discusses the implementation of Advanced Driver Assistance System on Unity game engine in order to validate the results of a usability study conducted in 2005 on a closed course system using BMW X-5 in a Virtual Environment. Our hypothesis is that there is fewer than 5-meter difference between initial human braking and initial autonomous braking in slowing to a stop from 65 miles per hour in simulated driving conditions.

Keyword: autonomous vehicles, virtual simulator, unity

1 Introduction

Because autonomous vehicle technologies are relatively new, there are large gaps in the literature as it relates to the usability of these systems. Where the usability has been studied, the bulk of this research has been conducted in controlled laboratory settings using a variety of simulation technologies.[Young 1997] In 2015, to investigate an increasingly common advanced driver assistance system, a usability study was conducted on a closed course using a 2015 BMW X-5. The study focused on the usability of the BMW's active cruise control system - a semi-autonomous vehicle technology that enables a vehicle, at highway speed, to slow in stop-and-go traffic and to come to a stop if necessary. The study used BMW's marketing materials as a conceptual framework; testing the system under the conditions that the manufacturer indicated the system was designed to be used. In short, the vehicle is supposed to slow down and stop for you. The results of our investigation were surprising however. The BMW was found to initially slow down at a consistent 80 meters from a preceding vehicle or object – a full 32.89 meters shorter than that of the human drivers in the study. Given the prevalence of simulator based research in this domain we surmised that perhaps human braking in a simulated environment informed the implementation of the production system (e.g. human braking in a simulator is equivalent to the braking of the BMW active cruise control system). Our research goal is to gauge participant performance using the ACC to be evaluated against established driver safety guidelines and heuristics. In Our hypothesis is that there is fewer than 5-meter difference between initial human braking and initial autonomous braking in slowing to a stop from 65 miles per hour in simulated driving conditions. By the end of our experiment, we reject our hypothesis after data analysis.

2 Background

While their specific implementations vary, Adaptive or Active Cruise Control (ACC) systems generally adjust a vehicle's speed to maintain a preset distance from a slowing preceding vehicle at highway speeds (55+ MPH). The main study conducted at the Project Green test facility in Greenville, South Carolina, was designed to evaluate key issues regarding the usability and operation of ACCs generally.[Brinkley 2014] Due to good existing relationship between Clemson University, BMW of North American and it's local South Carolina dealers, a 2014 BMW X-5 was used within the study. While the study evaluated the BMW ACC system with "stop and go" function specially, we believe that the – findings are largely generalizable to the majority of active cruise control systems with similar capabilities. The slowing distance of autonomous braking was on average approximately 22 meters shorter than that of human drivers. Earlier we planned to create the simulation on OpenDS and referred to the work by Green [Green 2014]. But due to very less documentation and the paid nature of tutorials we shifted to Unity3d to create and conduct our simulation due to the simulation work by other people [Chan 2015] and great support within community. Unity is a game development engine combined with an integrated development environment (IDE) that is capable of creating applications for Windows, Mac OS X, Linux, and other mobile platforms. Using the Unity engine, developers can position, scale, and add realistic physics to objects in the game system which enabled us to create and conduct the experiment on simulation. We used Logitech g920 which has a steering wheel and pedals which closely resemble those in passenger vehicles.[Paul Green 2014] For Data Analysis, we used Shapiro Test for testing the sample for normality [Shapiro 1965]. Paul Green, from University of Michigan Transportation Research Institute (UMTRI), has talked about how to improve the data collected by simulator as sometimes the system can be too steady among other possible reasons. [Green 2005]

3 Experiment

The experiment utilized a within-subjects design with initial braking distance as a dependent variable and braking type as an independent variable with two levels (autonomous braking and human braking). Within Subject design was chosen as we wanted to conduct the experiment on same people and compare their results. Driver with a detailed explanation of the purpose of the study and an informed consent document. Willing participants that participation can cease at any time. After obtaining consent, information regarding the function and operation of the system were provided. After driving practice the MDSI was be administered followed by a short period of rest. The Multidimensional Driving Style Inventory (MDSI) [3] is a 44 item questionnaire that identifies eight types of driving behaviors: (1) dissociative, (2) anxious, (3) risky, (4) angry, (5) high-velocity, (6) distress reduction, (7) patient, and (8) careful. As per our hypothesis, it was not required to analyze data from MDSI. At the start of the scenario participants were read aloud the task description from the printed copy. Two scenarios were simulated: (1) when the autonomous braking is applied by itself when the car is less than 80 meters of distance from the car in front and (2) when the brakes were applied by the driver.

Figure 1: Experimental Setup with Logitech g920 on left and image of simulation screen on the right which shows braking distance between two cars during simulation on top left of screen

4 Data Collection

We used unity for the simulation of the experiment. We had setup a laptop with the unity simulation and Logitech g920, a driving force game steering wheel, was used to control the car for braking, acceleration and steering wheel control which closely resemble those in passenger vehicles. Ten participants were recruited from graduate students, affiliated with the Harris UX lab along with the students in this course. Participation was limited to adult users who indicated an absence of motor skill or visual challenges that require specialized driving equipment. The experiment utilized a within-subjects design with initial braking distance as a dependent variable and braking type as an independent variable with two levels (autonomous braking and human braking). Apparatus - Two simulations were developed in the Unity game engine. In the first scenario the participant is asked to accelerate the simulated vehicle to 65mph and to *allow the vehicle* to slow when it encounters a preceding vehicle. In the second scenario the participant is asked to accelerate the simulated vehicle to 65mph and to decelerate *manually* to avoid a collision with the preceding vehicle. Procedure – After obtaining consent data collection was conducted in three parts: 1) Demographic form that collected participant age and sex 2) The pre-experimental MDSI and 3) Experimental task data.

5 Data Analysis

We logged on 10 values of the Initial Braking distance, which is the final distance between the participant's car and the next car in front after either the breaks are applied itself in Active Cruise Control (acc)in the 1st case and the participant itself applies break in the 2nd case which we also refer to as human braking condition (hbc). According to our experiment, the participant should break after being at 80 meters distance from the next car in front. We also logged on a binary value of whether a collision took place or not. Mean initial braking distance in simulated driving in the human braking condition, 47.2 meters, was 32.8 meters shorter than autonomous braking (80 meters) and The Standard Deviation was 11.380, as can be seen in figure 1 and 2. A T-test is a statistical examination of two population means. A single value t-Test showed a significant difference between mean human braking distance and the 80 meters of autonomous simulated braking with t-value of 9.1142 and df = 9 in our experiment. Mean initial braking distance in simulated driving in the human braking condition, 47.2 meters, was 65.69 meters shorter than that of participants in the naturalistic evaluation (112.89 meters)[5]. We also conducted the Shapiro Wilk test, which gave us w=0.94085 and p-value of 0.5625. So, since the p-value>0.5, we can accept the null hypothesis that data came from normal distribution. Finally, the difference between human braking condition and active cruise control is calculated for all the participants and its mean is 32.8 and standard deviation is 11.38. This value is more than 5, which rejects our hypothesis that that there is fewer than 5-meter difference between initial human braking and initial autonomous braking in slowing to a stop from 65 miles per hour in simulated driving conditions.

	Mean	SD	Min	Max
Human Braking Condition	47.2	11.38	39	71
Active Cruise Control	80	0	80	80

Table 1: Table Contains Mean, SD, Min, Max values for both braking conditions in our experiment

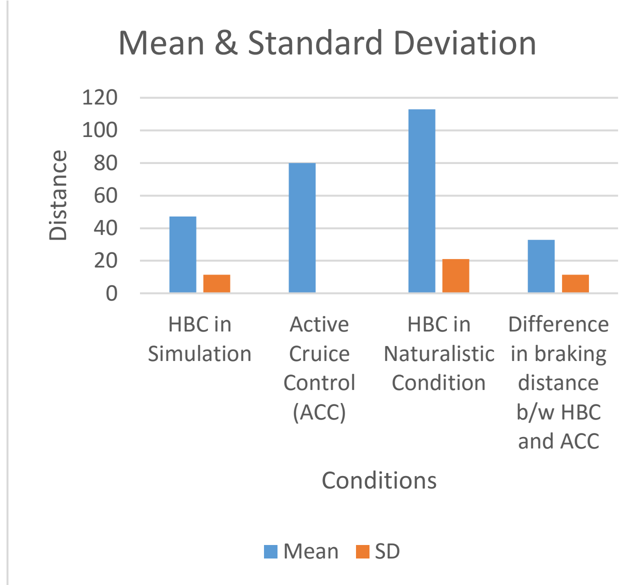

Figure 2: Mean and Standard Deviation for Human braking Condition and

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        Shapiro-Wilk normality test

data:  my.data[, 2]
W = 0.94085, p-value = 0.5625
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Figure 3: Shapiro-Wilk test for Human Braking Condition in our experiment

6 Conclusion

From figure 2 and 3, we know that our data is normally distributed and mean value of Human braking condition is lesser than active cruise control and our data is validated. After data analysis, we can see that our hypothesis of there being fewer than 5-meter difference between initial human braking and initial autonomous braking in slowing to a stop from 65 miles per hour in simulated driving conditions, is rejected. This paper presented the result of using virtual car simulation in place of an actual car for usability evaluation of the active cruise control system. Each scenario was explained, and its performance was analyzed by comparing the desired outcome. After the data analysis, we conclude that our hypothesis failed. While obvious variations exist between the simulated scenario and the exact conditions of the naturalistic evaluation, these results call into question the reliability of simulator-based studies when conducting usability evaluations of Advanced Driver Assistance Systems.

7 References

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